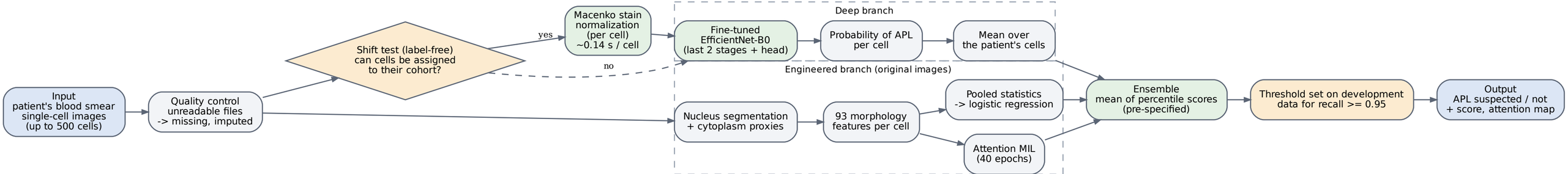
Input
patient's blood smear
single-cell images
(up to 500 cells)
Quality control
unreadable files
-> missing, imputed
Shift test (label-free)
can cells be assigned
to their cohort?
yes
no
Macenko stain
normalization
(per cell)
~0.14 s / cell
Deep branch
Fine-tuned
EfficientNet-B0
(last 2 stages + head)
Probability of APL
per cell
Mean over
the patient's cells
Engineered branch (original images)
Nucleus segmentation
+ cytoplasm proxies
93 morphology
features per cell
Pooled statistics
-> logistic regression
Attention MIL
(40 epochs)
Ensemble
mean of percentile scores
(pre-specified)
Threshold set on development
data for recall >= 0.95
Output
APL suspected / not
+ score, attention map